

Xinyu (Brian) Guo

FOUNDATION MODELS · GENERATIVE MODELING · AI FOR SCIENCE

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EDUCATION

Ph.D., Computational Biology & Bioinformatics University of Southern California · 3.95/4.0 · *Viterbi Fellow* 2022 – 2026

M.S., Biostatistics Johns Hopkins University · 3.97/4.0 · *Delta Omega Honor Society* Aug 2020 – May 2022

B.A., Mathematics & Computer Science Washington University in St. Louis · 3.89/4.0 · *Cum Laude* Aug 2018 – May 2020

EXPERIENCE

Postdoctoral Associate — Yale University Sep 2026 – Present

Foundation models & generative modeling · Advisor: Dr. Lucila Ohno-Machado · New Haven, CT

- Develop and train **generative flow-matching and diffusion models** that forecast how high-dimensional cell populations evolve over time; evaluate generalization to held-out time points and conditions.
- Fine-tune and evaluate **sequence foundation models** on long-context genomic data, including parameter-efficient (LoRA) adaptation and multi-GPU training on a Slurm cluster.
- Build **perturbation-response prediction models** for the Arc Institute Virtual Cell Challenge 2026; design leakage-safe, held-out-context benchmarks to measure out-of-distribution generalization.

AI Research Scientist Intern — Abbott Cancer Diagnostics (formerly Exact Sciences) Summer 2026

Genomic AI & precision oncology · San Diego (La Jolla), CA

- Evaluated sequence-to-function foundation models (AlphaGenome, Enigma) on an OncoExTra tumor cohort, expanding the reviewable evidence space by **2,554 of 6,087 variants (42%)** that snpEff had deprioritized.
- Built **OncoS2F**, an agent that turns natural-language requests into Enigma, Evo2, and Carbon workflows across **13 hosted genomic model endpoints**, from sequence construction through task-specific heads.
- Trained FFPE artifact-risk models on ~140 GB of paired fresh-frozen/FFPE sequencing, reaching **ROC-AUC 0.923** versus evaluated implementations of FFPErase (0.677), FFPolish (0.517), and FIFA (0.509) on the same cohort.
- Trained multiple-instance models on UNI2-h pathology embeddings over 2,099 whole-slide images, predicting ER (**AUROC 0.890**), TNBC (0.893), and HRD-high (0.790) with every leave-institution-out drop ≤ 0.03 .

Graduate Researcher — University of Southern California Aug 2022 – May 2026

Multi-modal representation learning · Advisor: Dr. Liang Chen

- Developed a **cross-modality foundation model** integrating DNA sequence, RNA expression, and spatial data, built for interpretability so individual inputs map to explainable spatial structure.
- Designed **self-supervised contrastive** GNN/GAT models with attention-based fusion to align population-scale genetic signals with spatial embeddings across single-cell modalities.
- Identified **trans-acting NMD-QTLs across 48 human tissues**, linking regulatory variation to nonsense-mediated mRNA decay and transcript stability.

Graduate Researcher — Johns Hopkins University Oct 2020 – Aug 2022

Statistical genetics · Advisors: Dr. Nilanjan Chatterjee & Dr. Diptavo Dutta

- Developed **CSTWAS**, a subset-based cross-tissue TWAS method that raises power over single- and multi-tissue tests while identifying which tissues drive each gene–trait signal.
- Ran transcriptome- and proteome-wide association studies at bulk and single-cell resolution, identifying genes linked to **renal cell carcinoma**.

Graduate Researcher — Johns Hopkins University Feb 2021 – Aug 2022

Multi-omics & cancer therapy resistance · Advisors: Dr. Hongkai Ji & Dr. Eneda Toska

- Built the lab's shared **RNA-seq, ChIP-seq, and ATAC-seq pipeline** and used it to show how the chromatin modifier **KMT2D** and its methylation by SMYD2 mediate hormone-driven therapy resistance.

Research Assistant — Washington University in St. Louis Jun – Aug 2020

Predictive clinical informatics · Advisors: Dr. Andrew Michelson & Dr. Philip Payne

- Shipped respiratory-failure prediction models for COVID-19 patients **into the Barnes-Jewish Hospital EHR**, running on live patient data to support ICU triage.

SELECTED PROJECTS

- Trained a **560M-parameter decoder-only LLM from scratch** 2025 · xinyuguo.com/projects/nanochat-llm-training — Rust BPE tokenizer, pretraining, supervised fine-tuning, and reinforcement learning with PyTorch DDP across 8 GPUs; evaluated on CORE, ARC, MMLU, GSM8K, and HumanEval.
- State-space sequence models 2025 – 2026 · lead developer — **TAL-Mamba**, temporal action localization over 2+ hour sequences at 68% less memory than Transformer baselines; **Track Mamba**, a linear-complexity real-time small-object tracker 40% smaller and 50% faster than TrackNet baselines.
- **Legal Patent Copilot** 2024 — retrieval agent for prior-art discovery over 10M+ documents using FAISS indexing and fine-tuned reranking layers, cutting search latency **42%** with improved recall.
- Benchmarked **AlphaGenome** against atrial-fibrillation TWAS 2025 · xinyuguo.com/projects/ag-twas-alphagenome — 10,120 fine-mapped variant–gene–tissue triplets across 10 GTEx tissues; chromatin tracks enriched for causal variants (ATAC 1.18×, $P < 10^{-4}$) while expression predictions held only 55–60% sign accuracy.

SELECTED PUBLICATIONS

1. [Guo, X.](#), Chen, L. From G1 to M: a comparative study of methods for identifying cell cycle phases. *Briefings in Bioinformatics* (2024).
2. [Guo, X.](#), Chatterjee, N., Dutta, D. Subset-based method for cross-tissue transcriptome-wide association studies improves power and interpretability. *Human Genetics and Genomics Advances* (2024).
3. Dutta, D., [Guo, X.](#), Winter, T.D., et al. Transcriptome- and proteome-wide association studies identify genes associated with renal cell carcinoma. *The American Journal of Human Genetics* (2024).
4. Bianchi, A., De Castro Silva, I., ... [Guo, X.](#), et al. Cell-autonomous Cxcl1 sustains tolerogenic circuitries and stromal inflammation via neutrophil-derived TNF in pancreatic cancer. *Cancer Discovery* (2023).
5. Blawski, R., Vokshi, B.H., [Guo, X.](#), et al. Methylation of the chromatin modifier KMT2D by SMYD2 contributes to therapeutic response in hormone-dependent breast cancer. *Cell Reports* (2024).
6. Kittane, S., Ladewig, E., ... [Guo, X.](#), et al. The histone methyltransferase KMT2D mediates subtype-specific transcriptional regulation, lineage plasticity, and therapeutic response in castration-resistant prostate cancer. *Cancer Research* (2024, 2025).
7. Jagirdar, K., Portuallo, M.E., ... [Guo, X.](#), et al. ERK hyperactivation as a unified mechanism of escape in intrinsic and acquired CDK4/6-inhibitor resistance in acral lentiginous melanoma. *Oncogene* (2024).
8. Dutta, D., Zhang, J., [Guo, X.](#), et al. Leveraging and partitioning polygenic risk scores to identify cancer-related proteins. *medRxiv* (2025, preprint).

TECHNICAL EXPERTISE

ML RESEARCH Foundation-model pretraining & fine-tuning (LoRA / full) · generative models (flow matching, diffusion, autoregressive LLMs) · transformers, state-space models (Mamba), GNNs · self-supervised & contrastive learning · reinforcement learning · agentic systems & tool use · RAG / vector retrieval · benchmark design & leakage-safe evaluation

ENGINEERING Python · C++ · SQL · Bash · TypeScript · R · PyTorch (DDP/FSDP) · JAX · Hugging Face · PyG · CUDA · distributed multi-GPU training on Slurm HPC and AWS (SageMaker, Bedrock, EC2, S3) · FastAPI · Docker · Git

AI FOR SCIENCE Genomic, single-cell, and pathology foundation models (AlphaGenome, Enigma, Evo 2, Borzoi, Enformer, scGPT, Geneformer, UNI) · variant-effect prediction · single-cell & spatial omics · statistical genetics